# PULSE WAVE OPERATION

## INTRODUCTION TO SEQUENTIAL LOGIC

*prepared by:*

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Electronics Engineer

# GRADING SYSTEM

# FINAL GRADE

30%

PRELIM

30%

MIDTERM

**40%**

ENDTERM

# TERM GRADE

50%

WEEKLY

50%

EXAM

# 100%

LABORATORY

# PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

## TOPIC OUTLINE

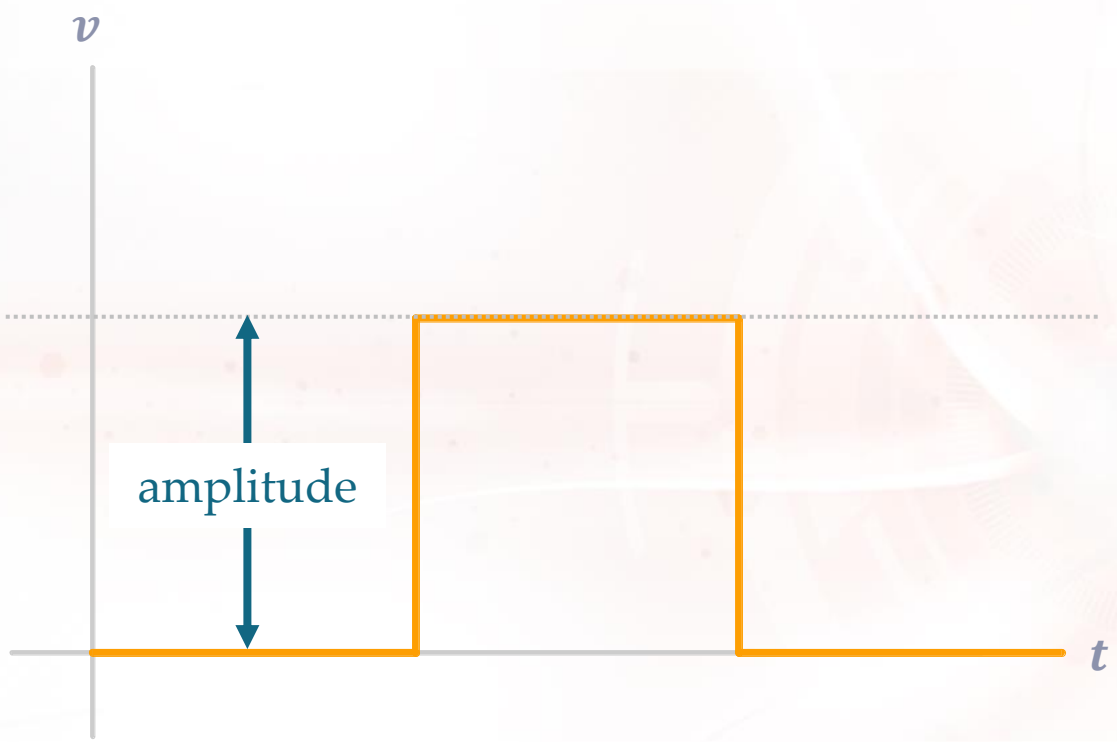
Pulse Wave Characteristics

Triggering Methods

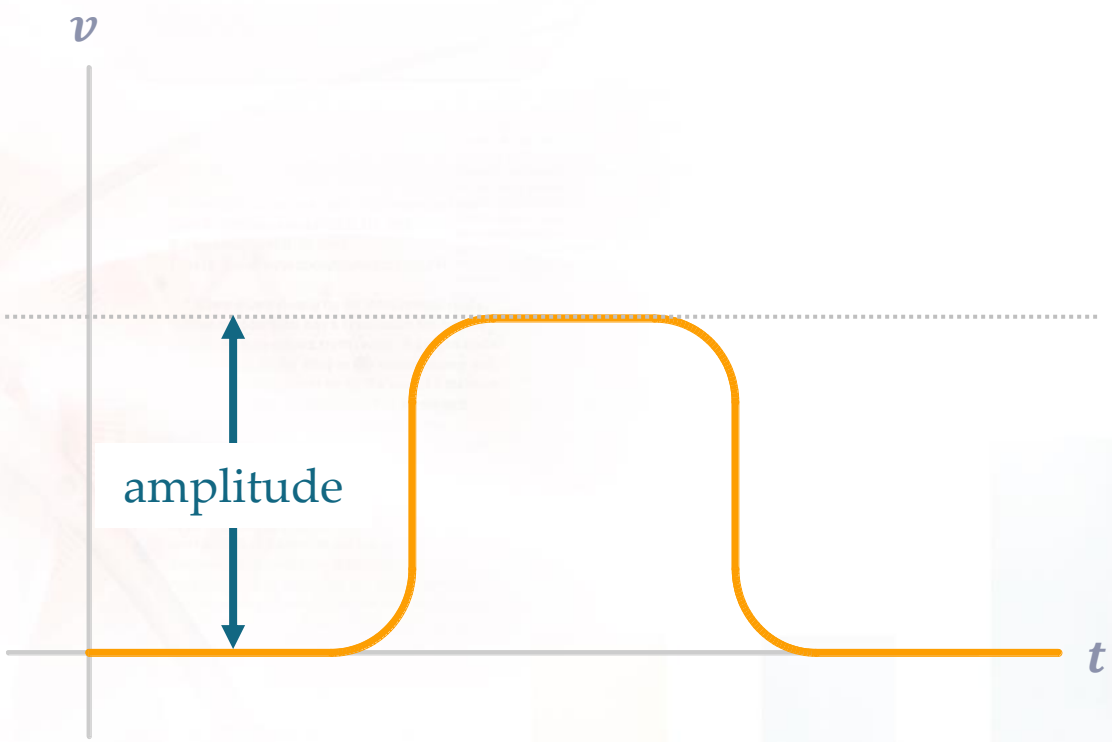
# PULSE WAVE CHARACTERISTICS

# AMPLITUDE

Ideal

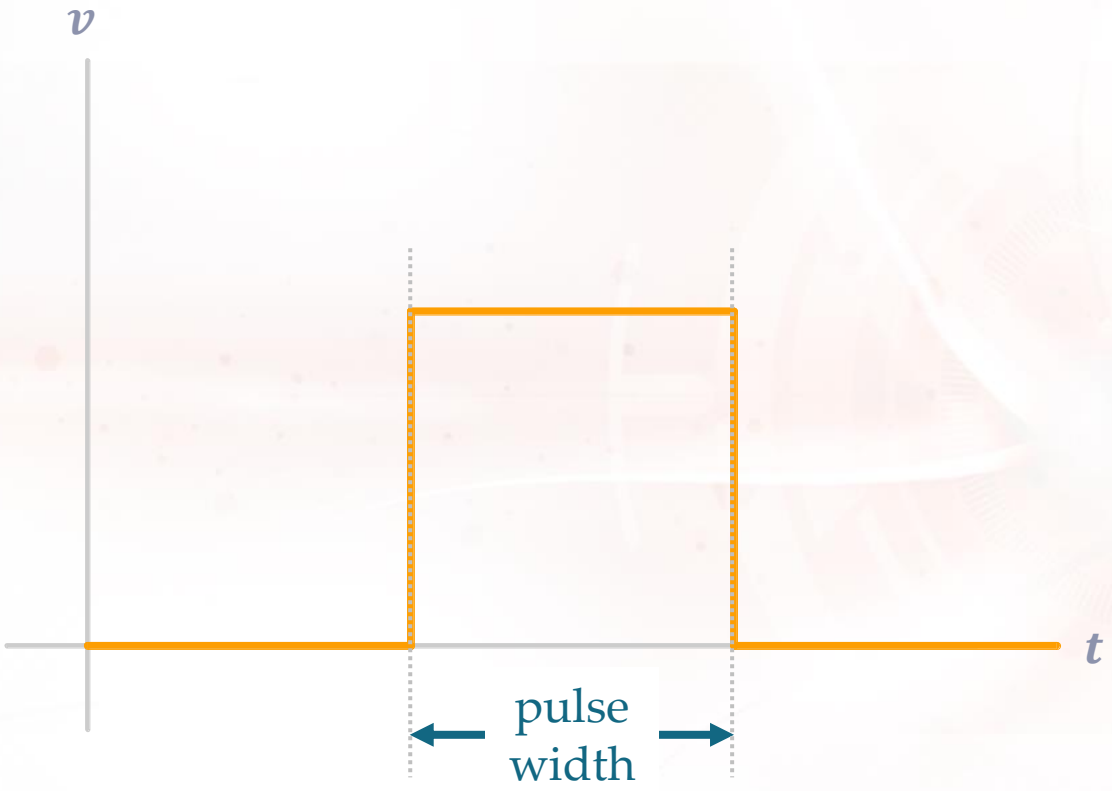



Actual

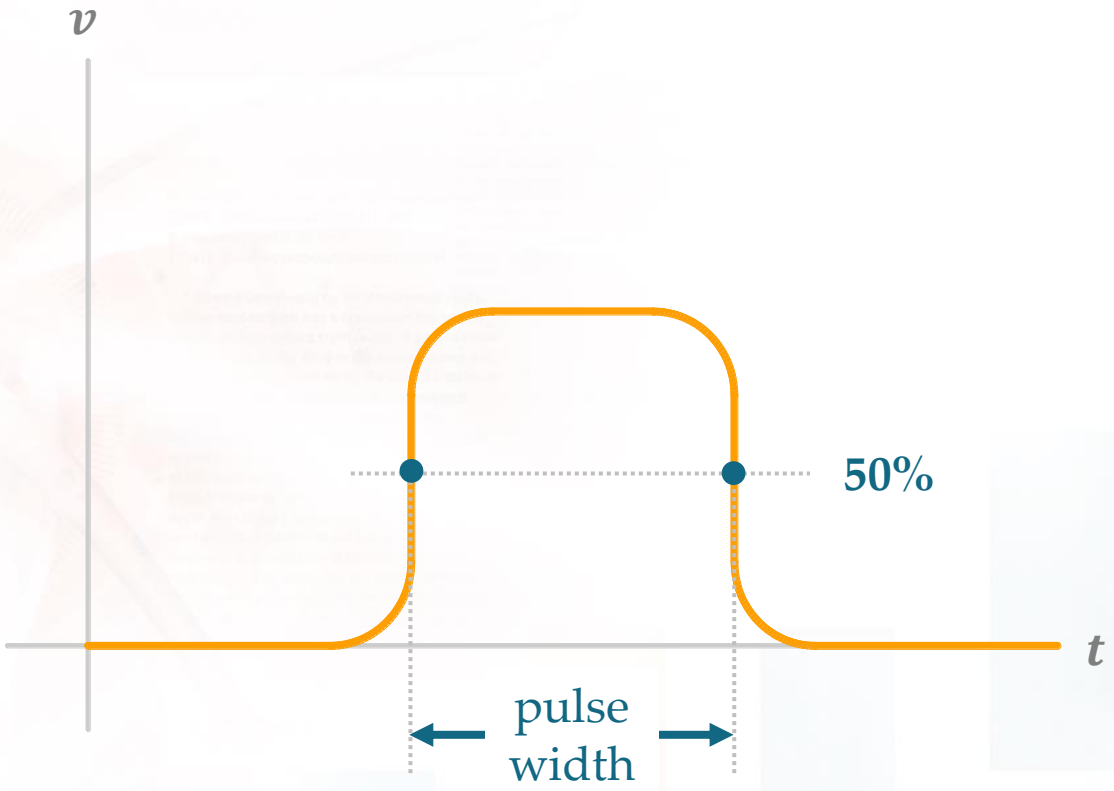


# PULSE WIDTH

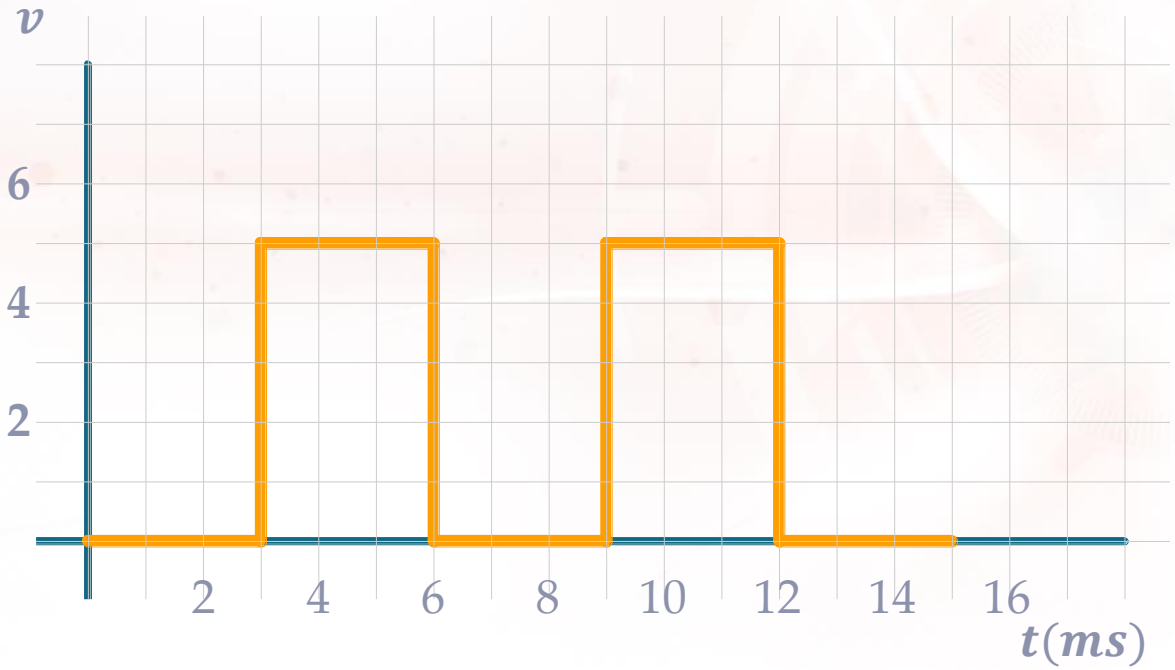
Ideal




Actual



Determine the **amplitude** and **pulse width** of the given **pulse wave**.



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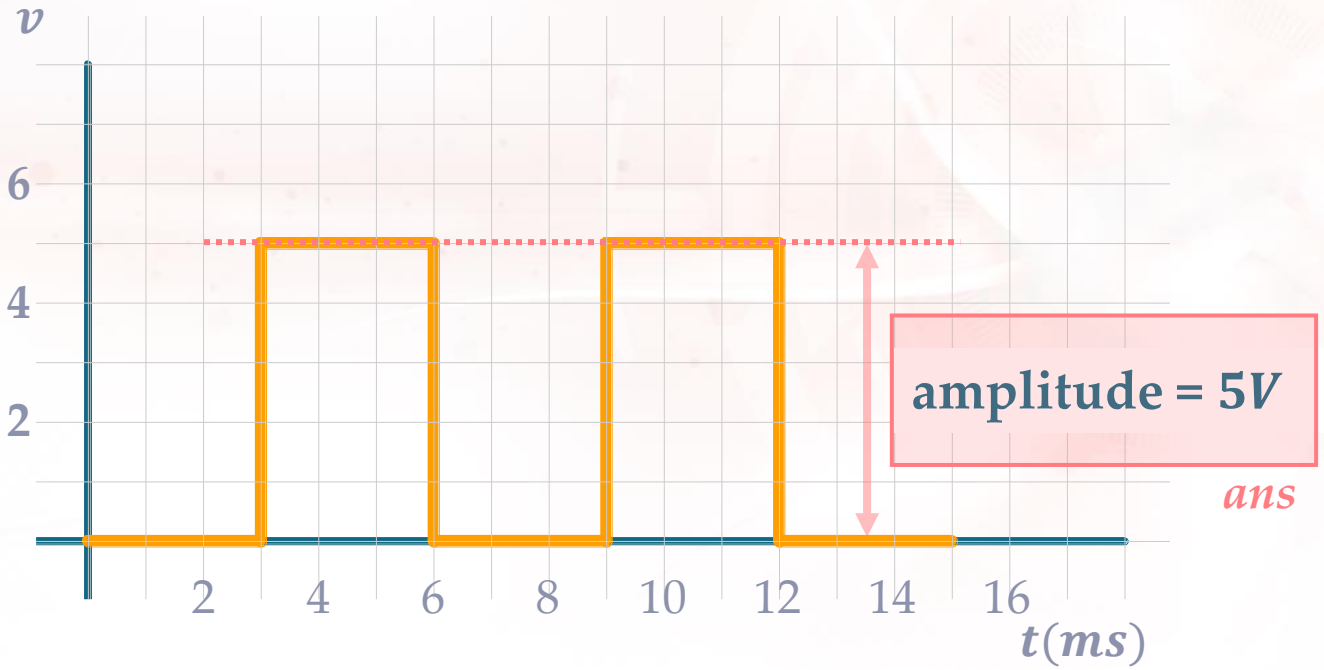
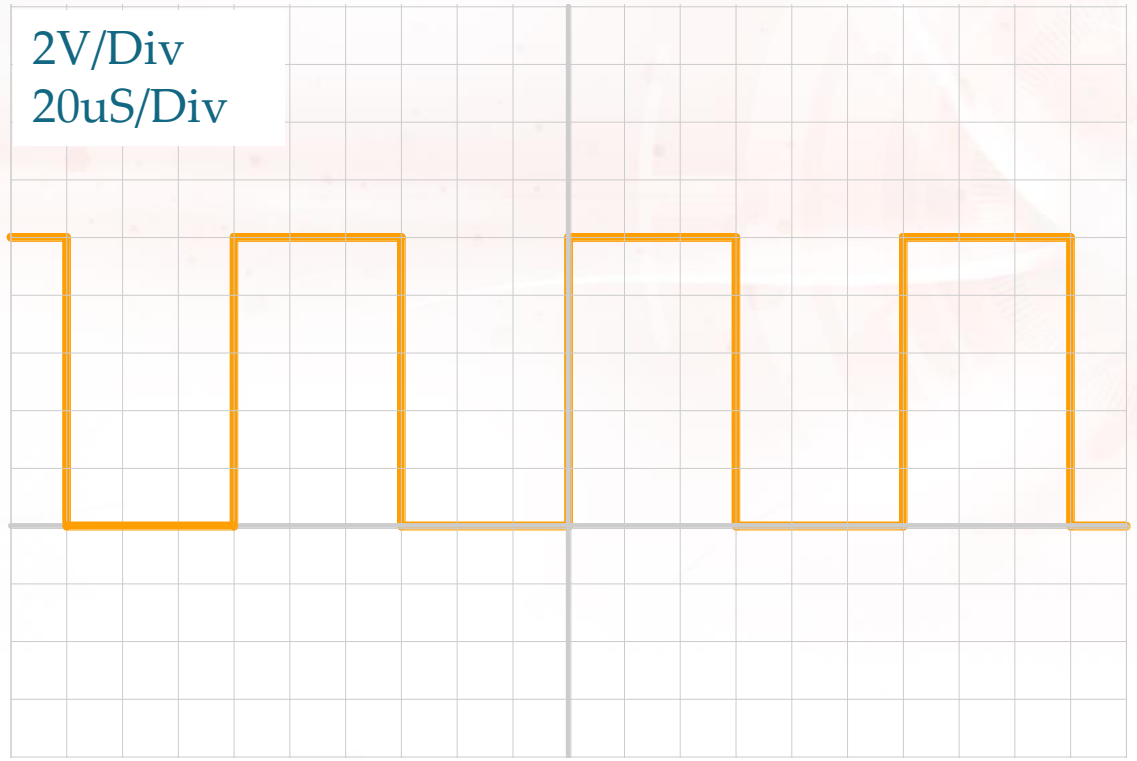


Determine the **amplitude** and **pulse width** of the given **pulse wave**.

**pulse width = 3 ms**

*ans*

Determine the **amplitude** and **pulse width** of the given **pulse wave**.

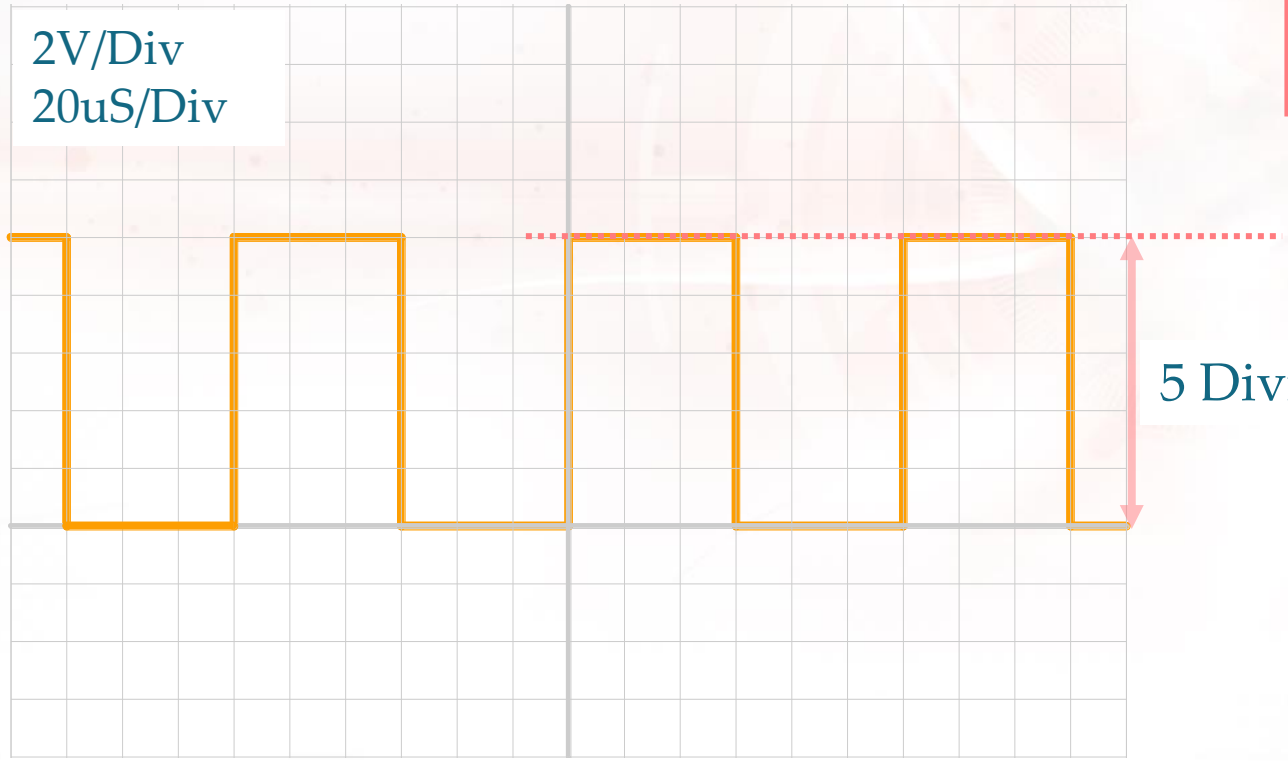


Determine the **amplitude** and **pulse width** of the given **pulse wave**.

$$\text{amplitude} = 5\ \cancel{Div\left(\frac{2\ V}{Div}\right)}$$

$$\boxed{\text{amplitude} = 10\ V}$$

*ans*



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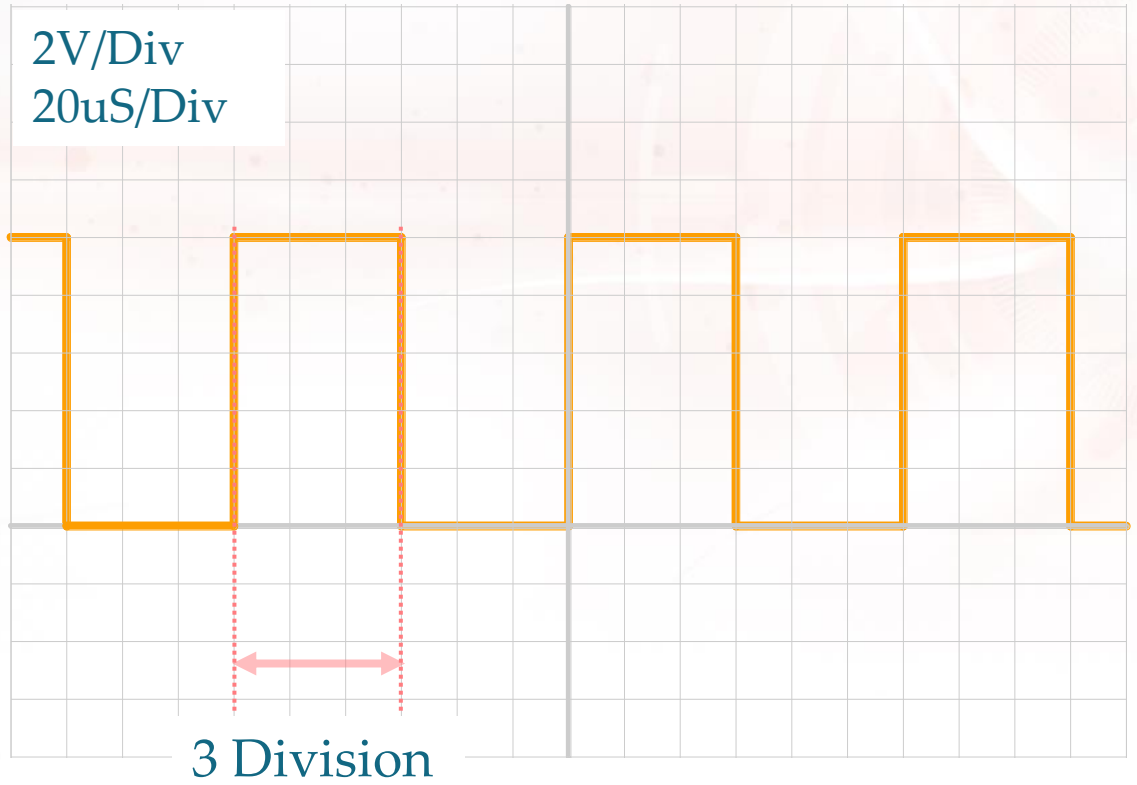


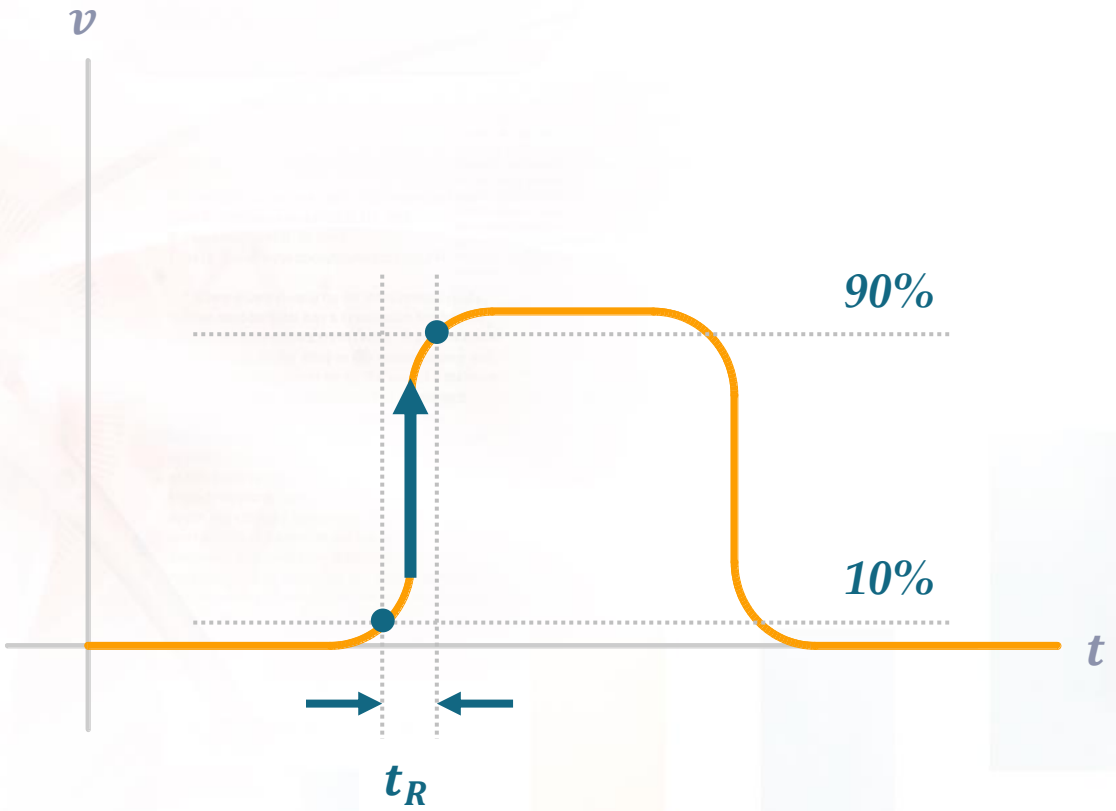

$$\text{pulse width} = 3\ \cancel{Div}\left(\frac{20\ \mu S}{\cancel{Div}}\right)$$

$$\boxed{\text{pulse width} = 60\ \mu S}$$

*ans*

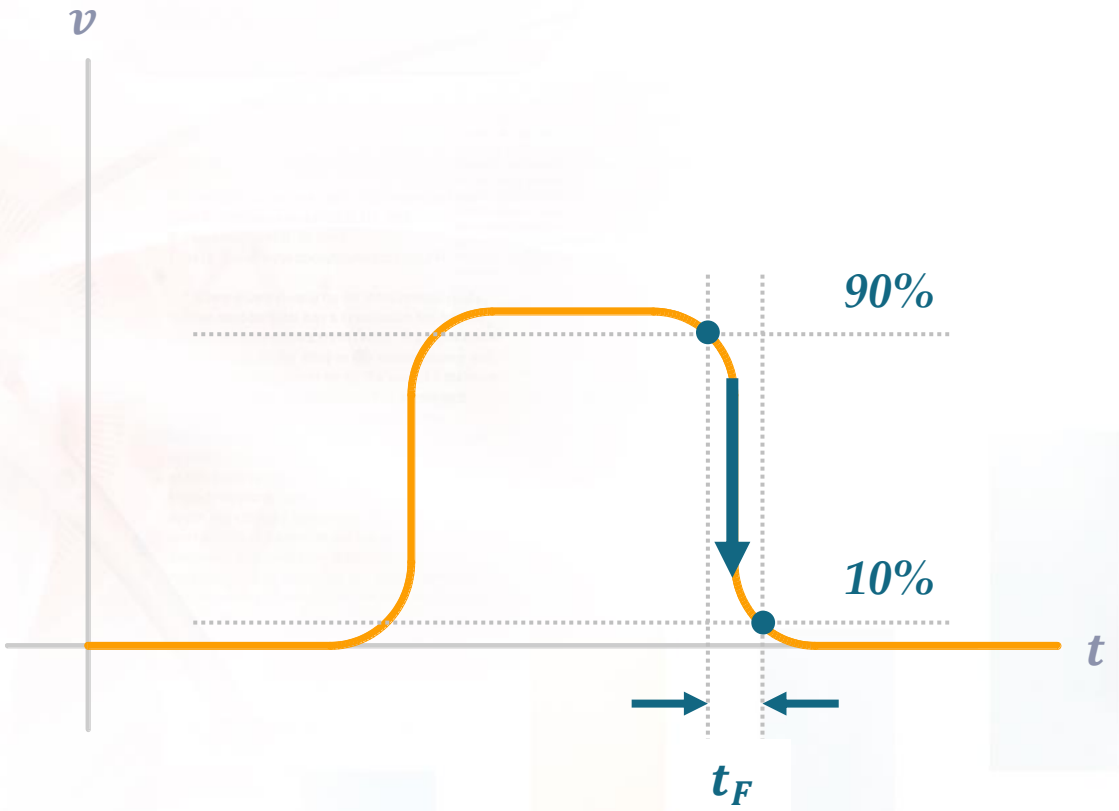
# RISE TIME

Time required for the **rising edge** of pulse to go from **10%** to **90%** of the amplitude



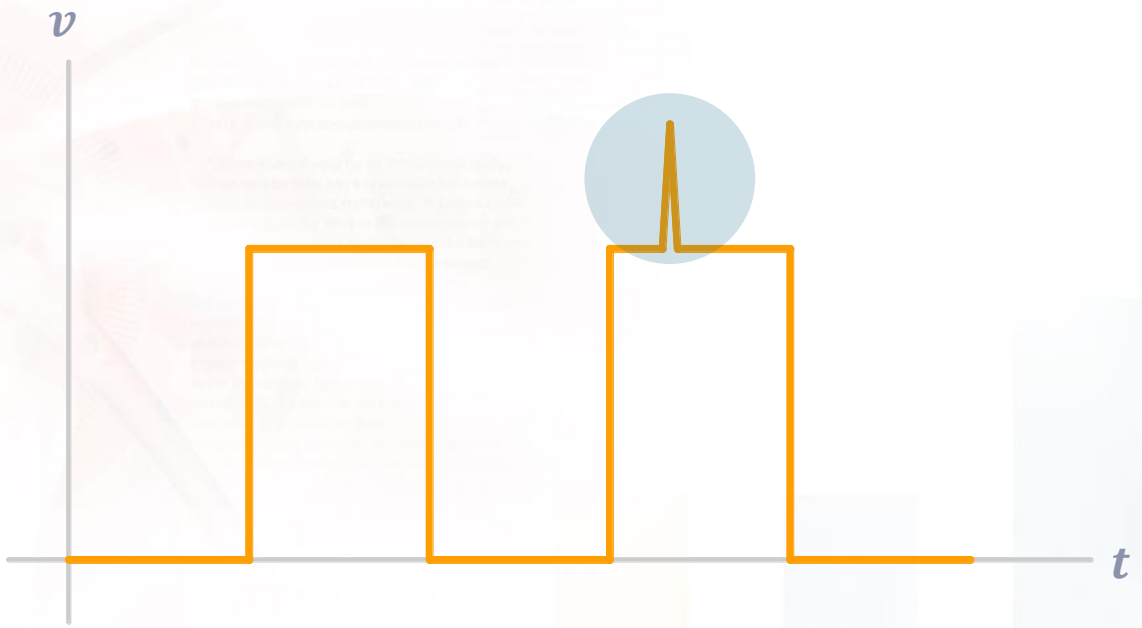
# FALL TIME

Time required for the **falling edge** of pulse to go from **90%** to **10%** of the amplitude



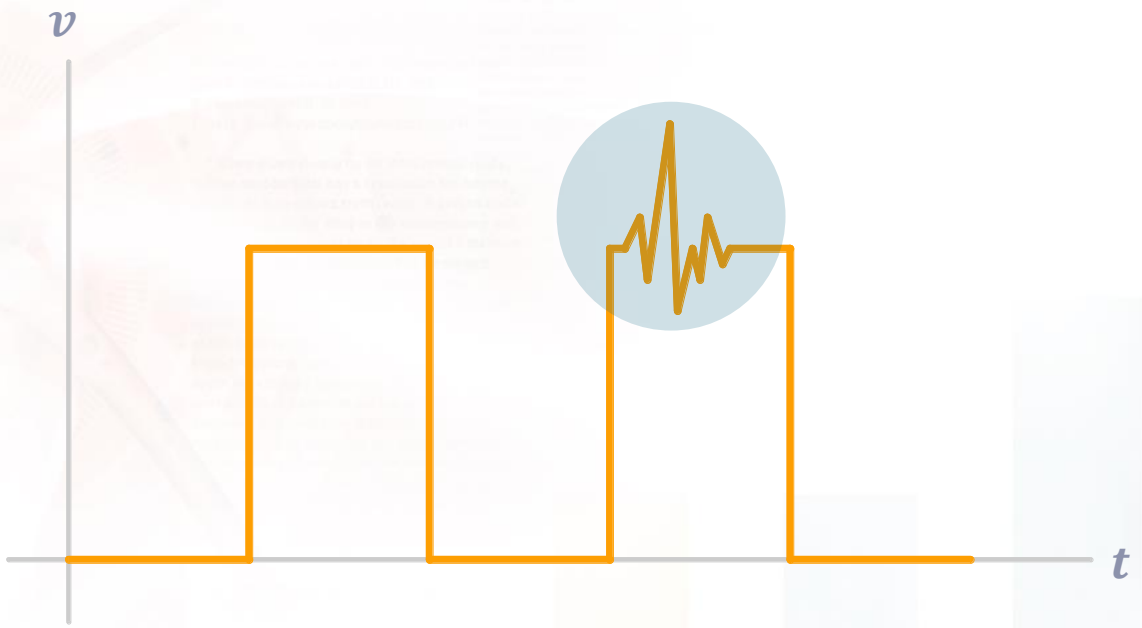
# OVERSHOOT

The waveform **exceeds** the desired value



# RINGING

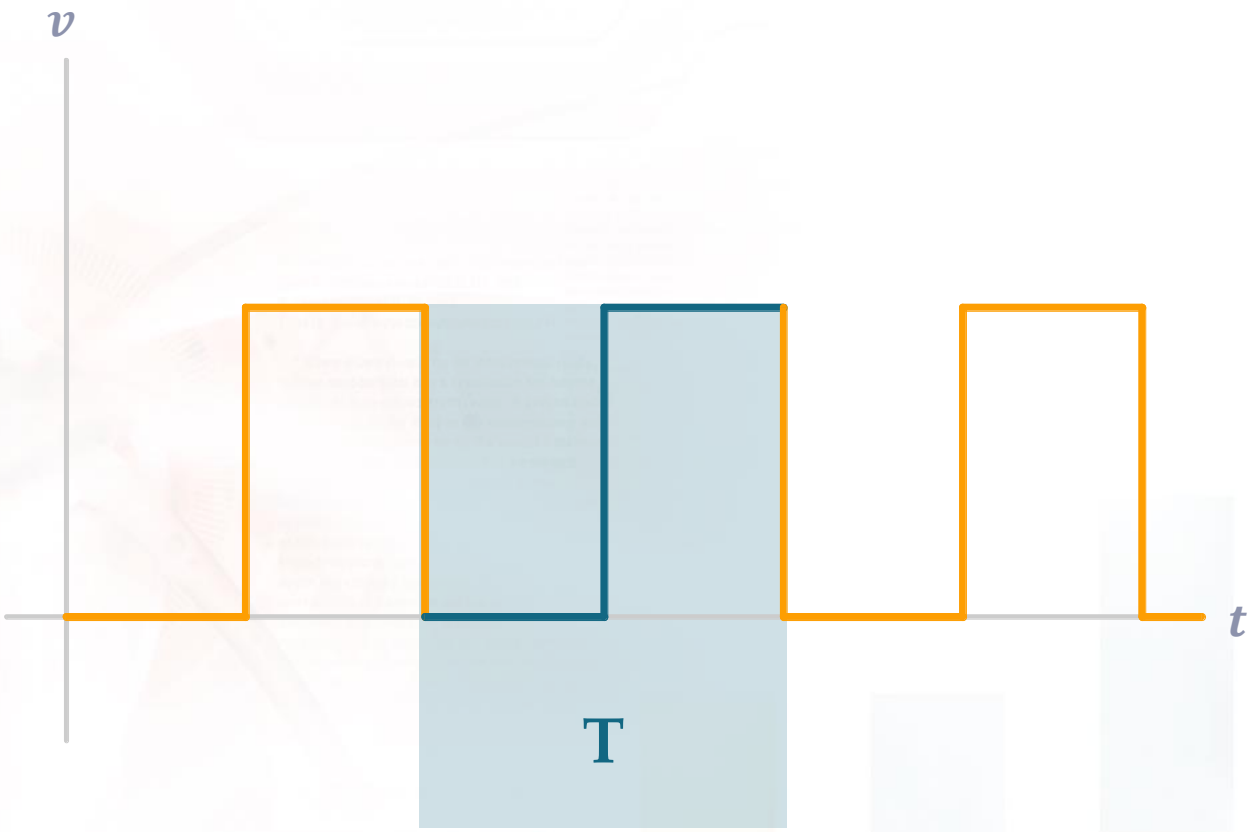
The waveform **alternates** about the desired value



# PULSE REPETITION RATE

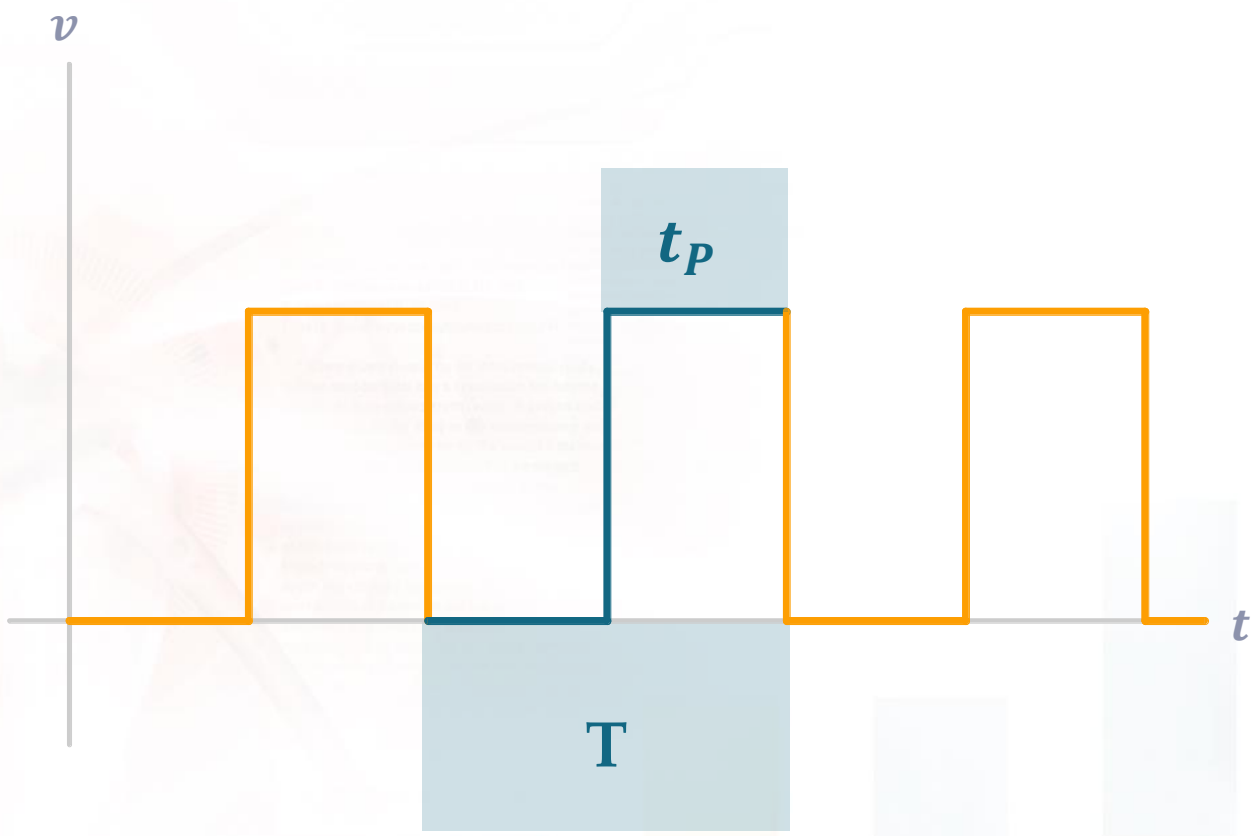
The **period** in which the pulse patterns repeats (pulse/second)

$$\mathbf{PRR} = \frac{\mathbf{1}}{\mathbf{T}}$$



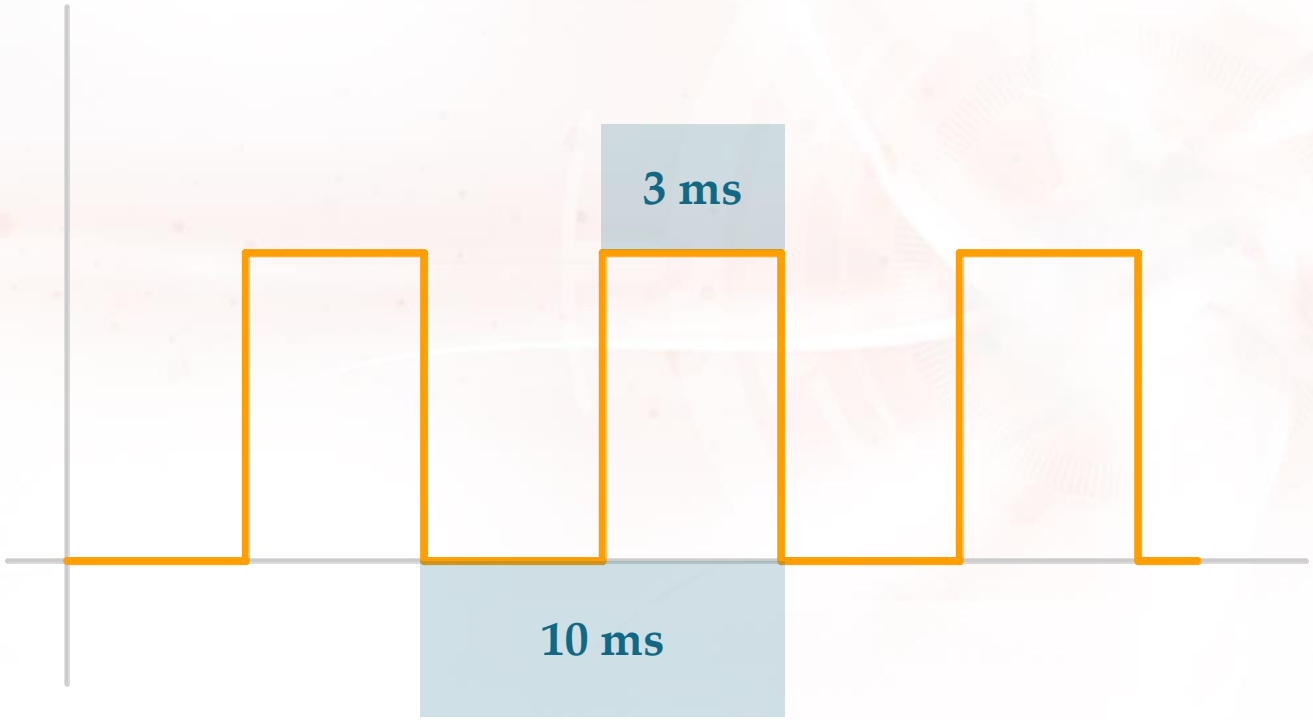
# PULSE DUTY CYCLE

Percent of the period that pulse is at its **high** level




$$\textbf{Duty Cycle} = \frac{t_P}{\mathrm{T}} \cdot 100\%$$

Determine the **pulse frequency** and **duty cycle** of the given **pulse wave**.



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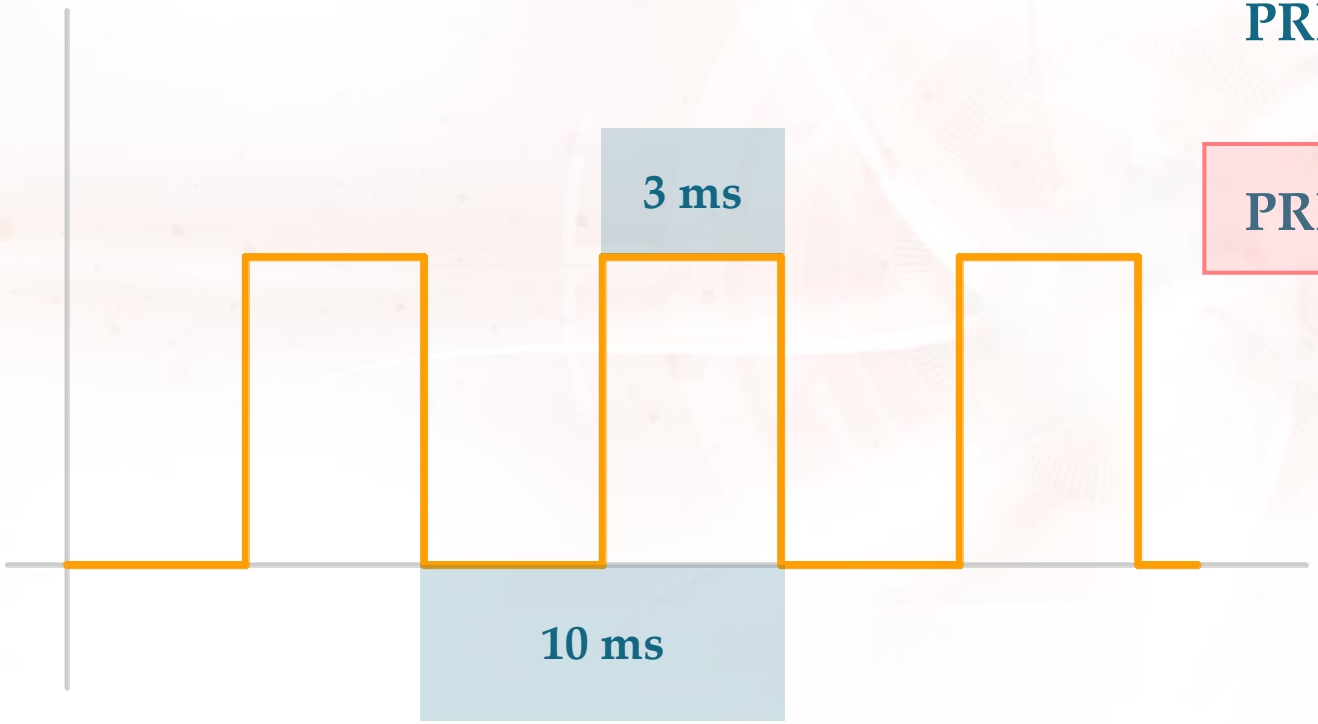


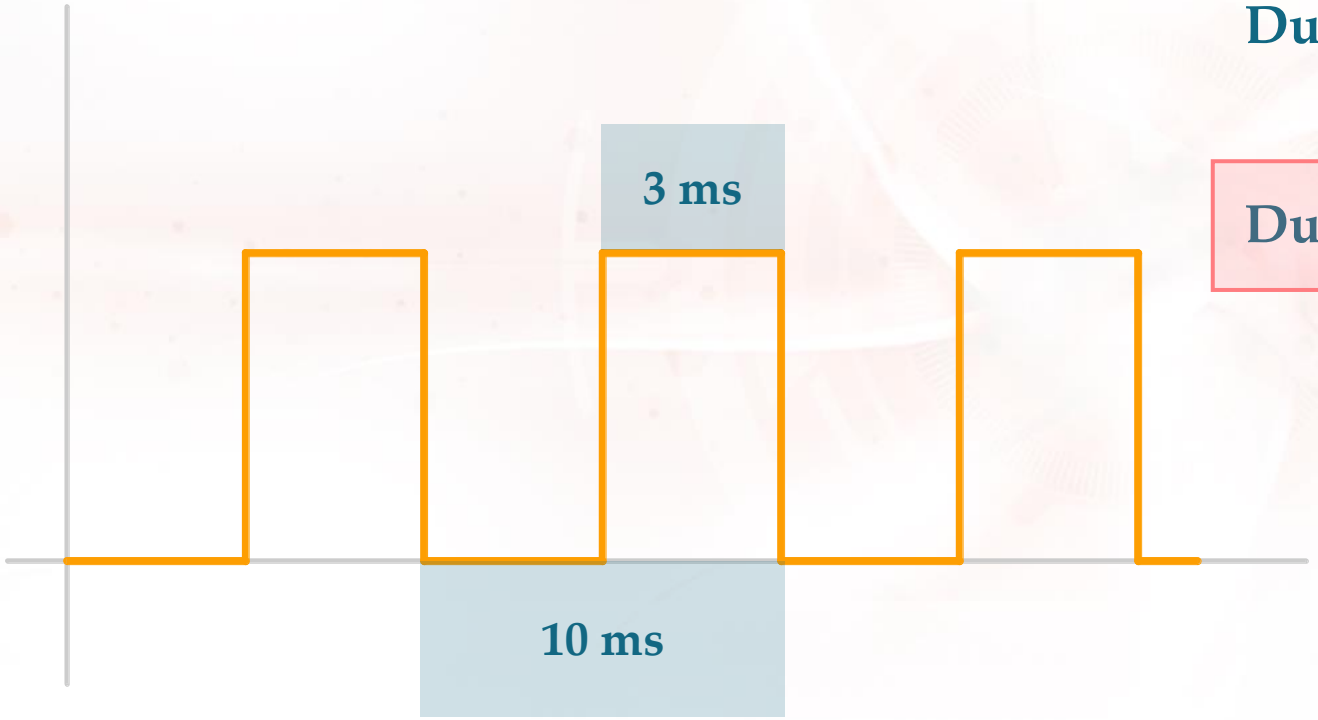

$$\text{PRR} = \frac{1}{T}$$

$$\text{PRR} = \frac{1}{10\ ms}$$

$$\boxed{\text{PRR} = 100\ Hz}$$

*ans*

Determine the **pulse frequency** and **duty cycle** of the given **pulse wave**.



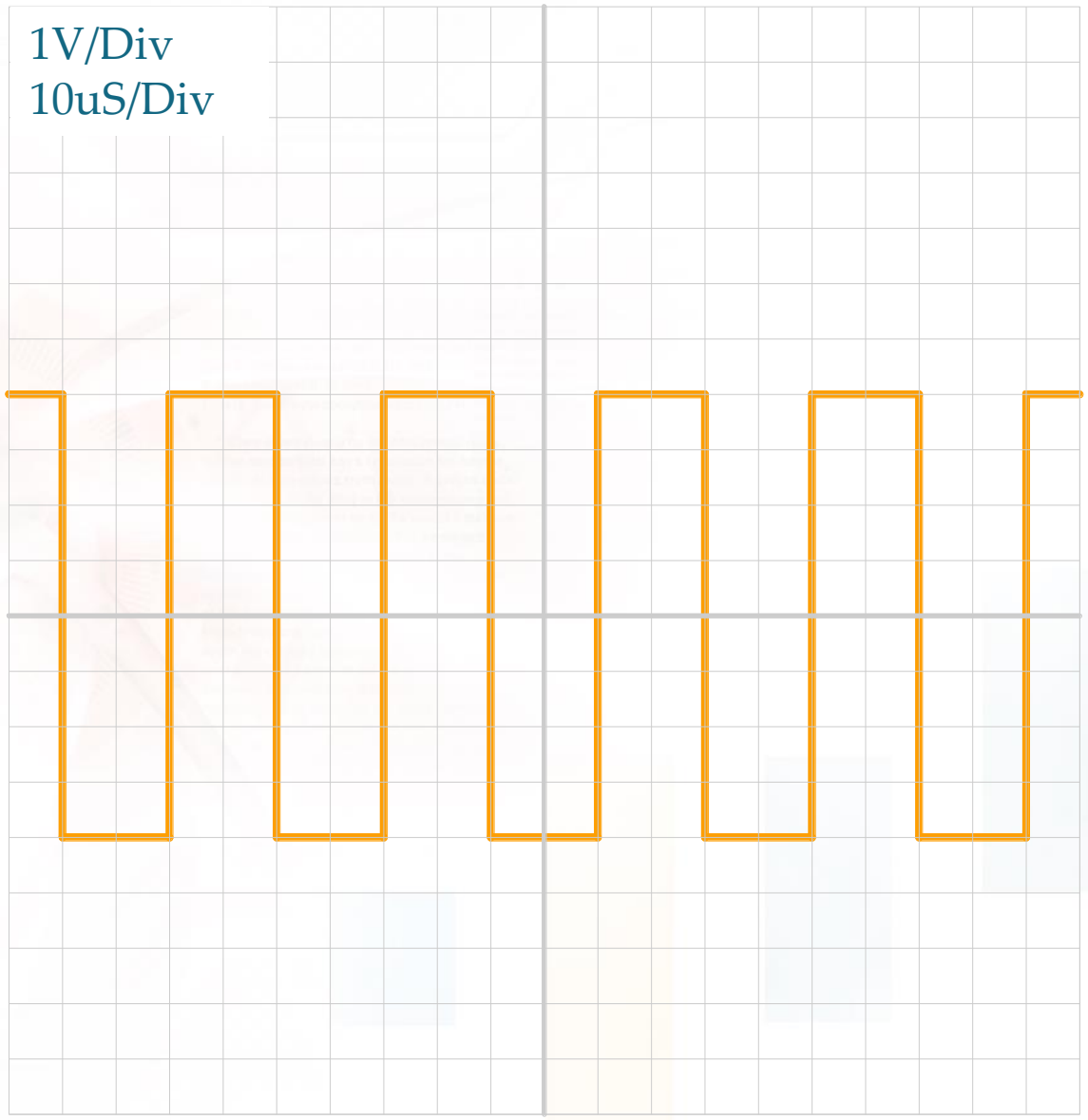

$$\text{Duty Cycle} = \frac{t_P}{T} \cdot 100\%$$

$$\text{Duty Cycle} = \frac{3\ ms}{10\ ms} \cdot 100\%$$

$$\boxed{\text{Duty Cycle} = 30\ \%}$$

*ans*

Determine the **pulse repetition rate** and the **duty cycle** of the given **pulse wave**.



Determine the **pulse repetition rate** and the **duty cycle** of the given **pulse wave**.

$$T = 4\ \cancel{Div}\left(\frac{10\ \mu s}{Div}\right)$$

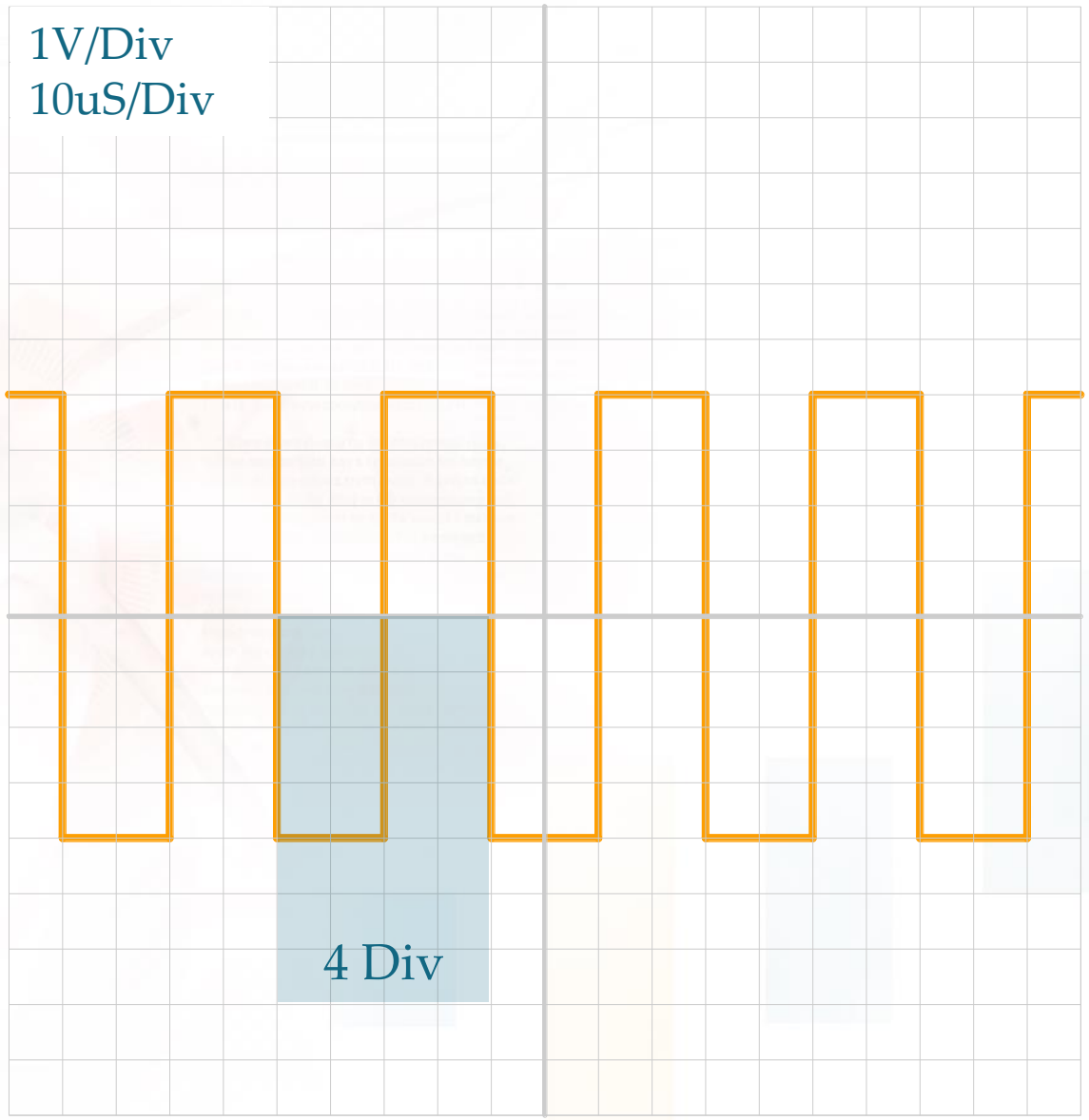
$$T = 40\ \mu s$$

$$PRR = \frac{1}{T}$$

$$PRR = \frac{1}{40\ \mu s}$$

$$\boxed{PRR = 25\ kHz}$$

*ans*



Determine the **pulse repetition rate** and the **duty cycle** of the given **pulse wave**.

$$t_P = 2\ \cancel{Div}\left(\frac{10\ \mu s}{\cancel{Div}}\right)$$

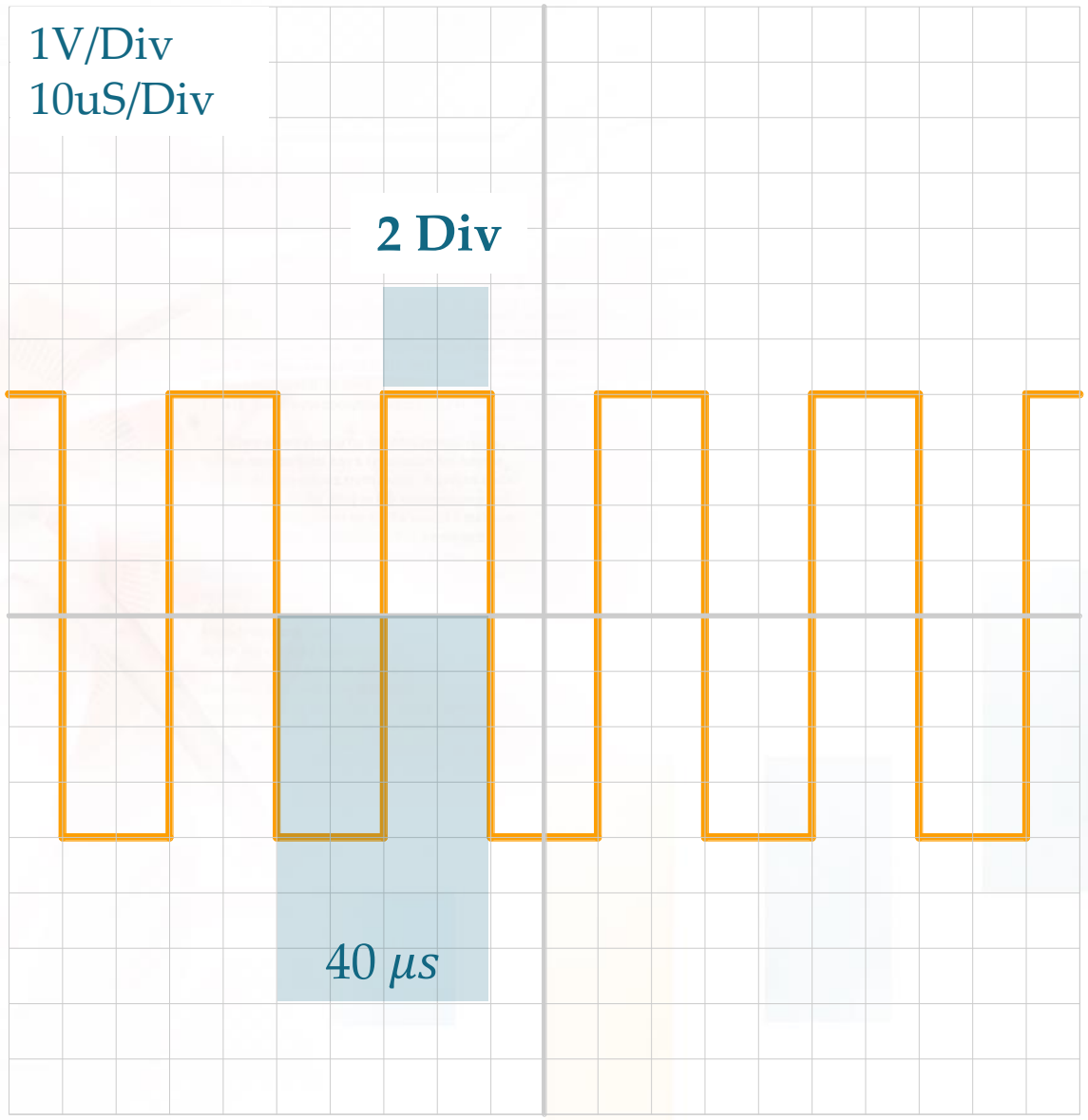
$$t_P = 20\ \mu s$$

$$\text{Duty Cycle} = \frac{t_P}{T} \cdot 100\%$$

$$\text{Duty Cycle} = \frac{20\ uS}{40\ \mu s} \cdot 100\%$$

$$\boxed{\text{Duty Cycle} = 50\ \%}$$

*ans*



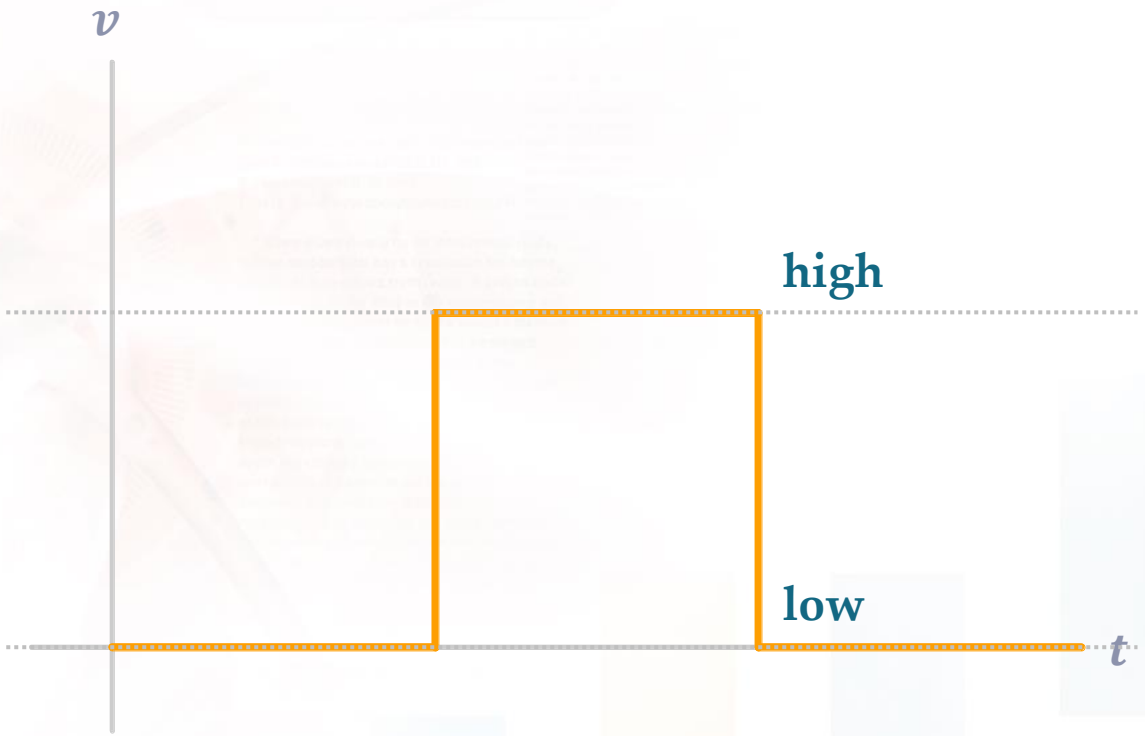
# TRIGGERING METHODS

# LEVEL TRIGGERING

Detection of a particular **signal level**

**Logic 1**

**Logic 0**

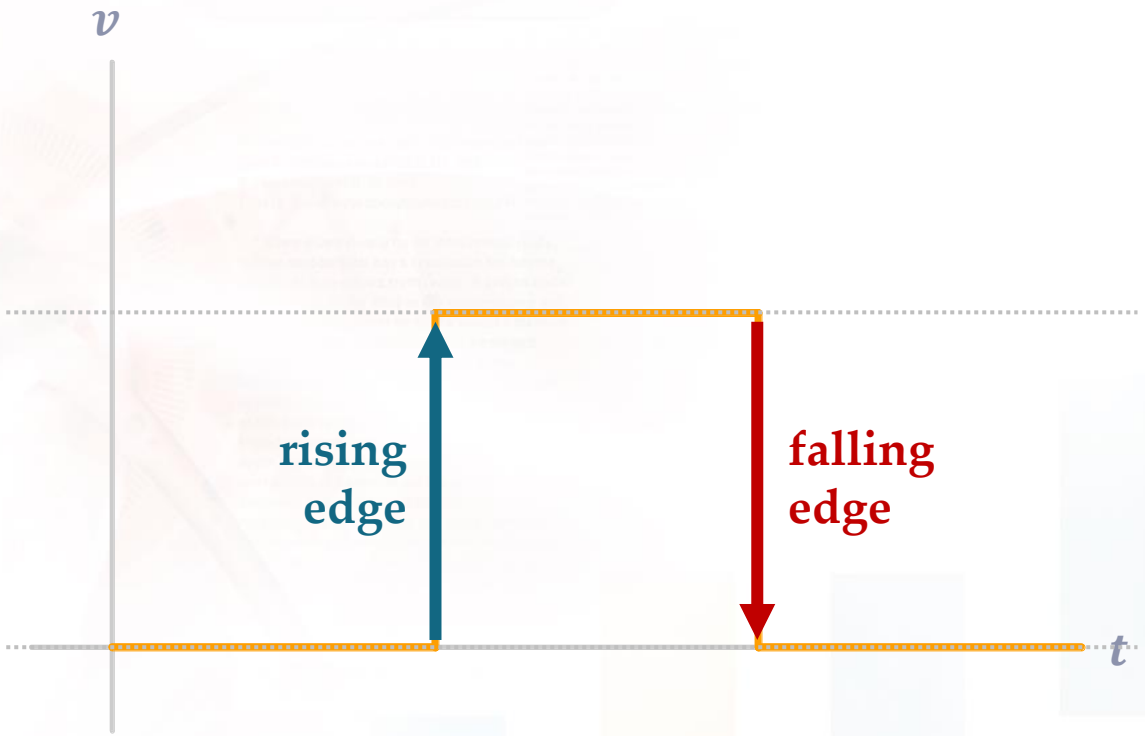


# EDGE TRIGGERING

Detection of a **change** in signal level

**0 to 1**

**1 to 0**



# LABORATORY